PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	The Global Prevalence, Treatment, and Outcome of Tuberculosis and COVID-19 Coinfection: A Systematic Review and Meta-analysis (From November 2019 to March 2021)
AUTHORS	Wang, Quan; Guo, Shasha; Wei, Xiaolin; Dong, Quanfang; Xu, Ning; Li, Hui; Zhao, Jie; Sun, Qiang

VERSION 1 – REVIEW

REVIEWER	Wilkinson, Robert Wellcome Centre for Infectious Diseases Research in Africa
REVIEW RETURNED	06-Feb-2022

GENERAL COMMENTS	A systematic review of published literature up until mid March 2021 on the possible clinical interaction between tuberculosis and SARS-CoV-2 is presented. The two questions appear to relate to coincidence and increased risk of death, and also to the treatments that were used. Conclusions are that most studies (35 reporting 55 cases) were case reports and, of the cohorts, only one (from South Africa) had really adequate power to test association. Defects in the quality of studies found was a frequent feature. The manuscript represents quite a lot of work and methodological rigor has been applied. As a review of case reports and cohorts alone the study has value although the extent two which meta-analysis was possible was quite limited. The following points are for the authors' consideration 1. The search is restricted to 6 databases. Preprints (unless captured) would be excluded. Ordinarily I would concur this was reasonable but so much literature on COVID-19 has gone into preprints that are yet to be published one wonders whether a search of that resource might have been justified here. Along similar lines the searches were run in mid-March 2021 which is nearly a year ago. Again, in the normal run of things this would be acceptable, but one suspects greatly more literature would have been found were the searches to be re-run. It may however hardly be reasonable to ask the authors to do this. However, I do suggest the title be qualified with 'up to March 2021' so that readers know the reference period. 2. TB-COVID-19 co-infection should be spelt out early (i.e. in the abstract) as the definition which seems to have been applied which is the detection of both pathogens during a single health care encounter? I would exclude the case of COVID-19 in a patient with latent tuberculosis because there must have been literally hundreds of millions of those. 3. It is not clear what the observed mortality is compared to, perhaps

	because it is not always possible, especially with case reports. Ideally the mortality of COVID-19 alone, and tuberculosis alone for each setting should be known but that clearly is not going to be possible. Also as the authors acknowledge the data was of insufficient quality to allow adjustment for cofounding co-morbid conditions or even age and gender. It should be pointed out that the excellent Study by Davies et al. was able to do this and remains the only benchmark work on this subject. The study incidentally is cited as a preprint but should be cited as formally published. 4. The usual abbreviation for lower- and middle-income countries is LMIC 5. Given that a shortage of cohorts was identified it seems not logical to have excluded cohorts of less than 10 whilst including individual case reports? 6. In section 2.2 all search terms should be stated (i.e. do not use etc.). Alternatively refer to the list of search terms provided in the appendix. 7. It is a matter of style but I have would have discussed the quality of studies before presenting the analysis, not afterwards. 8. It is not completely clear to the reviewer how the numbers add up in Figure 1 and this Figure therefore might benefit a legend that explains this better. 9. It would be respectful to the inventors to give a citation for the use of Covidence 10. Spelling of pyrazinamide 11. Grammar use needs correction in parts and would be improved even by the available function in Microsoft Word.

REVIEWER	Kurniawan, Andree Pelita Harapan University, Internal medicine
REVIEW RETURNED	03-Mar-2022

GENERAL COMMENTS	The authors have evaluated the Tuberculosis and covid 19 co-infection. My comments are 1. The relation between tuberculosis and covid in the introduction should be added that may relate to post tuberculosis or severe tuberculosis infection that compromises the lung capacity. 2. The included studies were comprised of retrospective studies and case reports. How the evaluation of quality studies? good? fair? bad? Only good quality of study could advance to meta analysis. 3. In the discussion should discuss the limited current evidence between Tuberculosis and COVID 19, further large prospective cohort studies are needed. 4. The conclusion was to overestimate the results. The conclusion: there was still limited evidence to evaluate the association between tuberculosis and covid 19. Further prospective cohort studies are needed.

VERSION 1 – AUTHOR RESPONSE

Dear Dr. Wilkinson,

On behalf of all the authors, I would like to express our sincerest thanks for your kind review. We believe your comments are very positive and valuable. All the changes for your comments in our manuscript have been highlighted by red text. Here below we address the questions and suggestions.

No.	Comment	Response
1	The search is restricted to 6 databases. Preprints (unless captured) would be excluded. Ordinarily I would concur this was reasonable but so much literature on COVID-19 has gone into preprints that are yet to be published one wonders whether a search of that resource might have been justified here. Along similar lines the searches were run in mid-March 2021 which is nearly a year ago. Again, in the normal run of things this would be acceptable, but one suspects greatly more literature would have been found were the searches to be re-run. It may however hardly be reasonable to ask the authors to do this. However, I do suggest the title be qualified with 'up to March 2021' so that readers know the reference period.	Thanks for your considerate comment. It is very challenging for us to add more database at this stage. When we designed this study, we were not sure about how many papers would be included in our study. As you mentioned, the preprint databases were not required for systematic review. Besides, we were not sure about quality of preprint studies since they were not under peer review. I agree the TB-COVID coinfection is a very new topic and for the new topic, preprint studies could be very valuable. Actually, we have designed another systematic review about this topic which will be conducted at latter half of the year, and I will definitely check the preprint databases. We have refined the tittle as you suggested. The new title is "*The Global Prevalence, Treatment, and Outcome of Tuberculosis and COVID-19 Coinfection: A Systematic Review and Meta-analysis (Till to March 2021)*". **As shown in P1.**
2	TB-COVID-19 co-infection should be spelt out early (i.e. in the abstract) as the definition which seems to have been applied which is the detection of both pathogens during a single health care encounter? I would exclude the case of COVID-19 in a patient with latent tuberculosis because there must have been literally hundreds of	Till to now, I'm afraid the TB-COVID coinfection is still not a widely-used term with specific definition. Most studies only use the word coinfection without clarifying what it really is. Therefore, I add the definition in abstract as "TB and COVID-19 (TB-COVID) coinfection means the disease caused by both mycobacterium tuberculosis and SARS-COV-2 infection." Besides, I also discussed the absence of definition at introduction part as following:

	millions of those.	*Although cases of coinfection and studies about prevalence status have been reported from several countries, we did not find a common definition about the TB and COVID-19 (TB-COVID) coinfection. In this systematic, we preliminarily described the TB-COVID coinfection as the disease caused by both mycobacterium tuberculosis and SARS-COV-2 infection.* **Please check P1-2 and P5.** I agree the number of latent TB and COVID-19 patients must be incredible huge, and I also understand the latent TB patient could be quite different with active ones. While based on the data I collected, only 1 out of 50 patients from the case reports were clearly described as latent TB patient. The readers of this systematic review could easily reach the basic information about this latent TB patient at Table S3 and S4 in appendix 3. There is very clear notice about this latent TB patient in the manuscript and the readers would not miss it (at P12). I think a possible explanation for why so few latent TB patient was reported is that doctors would not specifically to test the TB on COVID-19 patients. Besides, there has been study suspects the SARS-COV-2 might re-activate TB[1]. As for the possible latent TB patient included from 7 retrospective cohort studies, I' m afraid we just cannot tell them. For instance, the valuable study led by Dr. Davies provided numbers of previous tuberculosis patients and current tuberculosis patients. However, we cannot tell how many of them were latent TB. Of course, I can remove the only 1 patient with latent TB from this study, while I think it might be better to keep the patient since we do not know if there are interaction between the latent TB and COVID-19.

3	It is not clear what the observed mortality is compared to, perhaps because it is not always possible, especially with case reports. Ideally the mortality of COVID-19 alone, and tuberculosis alone for each setting should be known but that clearly is not going to be possible. Also as the authors acknowledge the data was of insufficient quality to allow adjustment for cofounding co-morbid conditions or even age and gender. It should be pointed out that the excellent Study by Davies et al. was able to do this and remains the only benchmark work on this subject. The study incidentally is cited as a preprint but should be cited as formally published.	Agreed! Theoretically speaking, our results, namely the overall and in-hospital coinfection fatality rates, should be compared with fatality rate of COVID-19 and TB alone. While it is not the aim of this study to estimate the fatality rate of TB or COVID-19. Therefore, we utilized the results from other study. We compared our results with Meyerowitz-Katz's study about the fatality rate of COVID-19. It necessary to point out that Meyerowitz-Katz's study included 36 highly heterogeneous studies whose samples contained COVID-19 patients could have other disease. However, I think that is the best study we can find. **Please check for P23 or as following:** *We tried to conduct MA on overall and in-hospital fatality rates, and the estimated results were 13.9% and 17.5%. According to Meyerowitz-Katz, the fatality rate of patients with single COVID-19 was 0.68% (95%CI: 0.053% ~ 0.82%), which was relatively smaller. Therefore, we believe the coinfected patients were facing more challenges and calling for the need to refine medical equipment and interventions. In addition, significant heterogeneity was detected in both MAs on overall and in-hospital fatality rates. In contrast, no heterogeneity was detected in both sub-groups' MAs on in-hospital fatality rate. We did not detect any pattern about treatment in both high-income countries and LMICs, based on case reports. Considering patients in high-income countries are likely to receive care of better quality than patients in LMICs, we guess the different results of MAs' heterogeneity might hint that the quality of care is highly associated with outcomes. In*

		other words, the patients in high-income countries shared a better condition than those in LMICs and were more likely to recover from TB and COVID-19 coinfection. It's necessary to point out that the results of MA were based on limited number of studies, of which defects in the quality was a frequent feature. The fatality rate of TB is more complicated. TB is more like a chronic disease which is rather different from COVID-19. I think the fatality rate of TB is highly related to the time. We do not know what time should we choose. Normally, it should be 3, 6, 9, 12, 18, and 24 months in the post-treatment phase. Even 3 months is far more than COVID-19's course of disease. Therefore, we did not compare the fatality rate between TB and coinfection. One thing I might need to point out. We only utilized the results from 7 retrospective cohort studies to estimate the fatality rate and Meta-Analysis. The results from case reports only provided us with information about medication and clinical outcomes. If may, I would like to express my sincere thanks for Dr. Davies and all other authors for their outstanding work in this field. I have cheeked the database and I find this paper is now published. All the citation has been renewed.
4	The usual abbreviation for lower- and middle-income countries is LMIC	Thanks for pointing out. I never noticed that. All the abbreviation has been renewed.
5	Given that a shortage of cohorts was identified it seems not logical to have excluded cohorts of less than 10 whilst including individual case reports?	There are 3 kinds of information we tried to collect: prevalence status, treatment and outcomes. The case report usually provides information about detail treatment, thusly we included them. For the cohort study, those provided any information about treatment or clinical outcomes (i.e. hospital stay, ICU) would be included in this systematic review, no matter how many

		its sample is. For cohort study only provides fatality rate, we excluded the ones with less than 10 patients. **Check Figure 1 in this letter for detail process.**
6	In section 2.2 all search terms should be stated (i.e. do not use etc.). Alternatively refer to the list of search terms provided in the appendix.	All the term has been added at section 2.2. *The searching process utilized Medical Subject Heading (MeSH) terms and Tittle/Abstract words around two areas: 1) COVID-19, including COVID-19, SARS-COV-2, 2) tuberculosis, including tuberculosis, TB, Mycobacterium tuberculosis, M.tuberculosis (see Table 2 in Appendix).* **Please check for P7.** Besides, list of search terms, the detail process of literature search in each database and the number of literatures found by each step are all showed at Table 1 in Appendix 2.
7	It is a matter of style but I have would have discussed the quality of studies before presenting the analysis, not afterwards.	We have replaced the analysis and quality of studies, as you suggested. **Please check for P21.**
8	It is not completely clear to the reviewer how the numbers add up in Figure 1 and this Figure therefore might benefit a legend that explains this better.	We have refined the description about data extraction process. **Please check for P10-11.** *We searched 565 studies from PubMed, 682 from ProQuest, 839 from Scopus, 644 from Web of Science, 367 from Embase, and 1 from the Cochrane database. After removing 1878 duplicate studies, we got 1220 studies and imported them into Covidence. With the help of Covidence, we excluded 1049 studies in the title and abstract screen step. Next, we carefully reviewed the full text of the remaining 171 studies and then we excluded 131 studies from them. We also found another 11 possibly relevant studies during the full text reviewing process, 9 of which were excluded. The most common reasons for exclusion were no outcomes of interest (n = 80) and no quantitative data (n=39).*
9	It would be respectful to the	Thanks for letting me know, and I will notice that in my

	inventors to give a citation for the use of Covidence.	further research. The citation has been added into our manuscript. **Please check for P8.** *Covidence (Covidence Company, Australia) was utilized to help us finish all the above processes.*
10	Spelling of pyrazinamide	The spelling of pyrazinamide has been corrected. **Please check for P15.**
11	Grammar use needs correction in parts and would be improved even by the available function in Microsoft Word.	I have refined the manuscript with help from Microsoft Word and Open Grammarly. I also have asked Prof. Xiaolin Wei, an associate professor in School of Public Health, University of Toronto, and Dr. Quanfang Dong, a postdoctor in the same institute to recheck the manuscript.

Dear Dr. Kurniawan,

Please accept our sincerest thanks for your kind reviewing on our work. Your summary and comments on our study are precise and valuable. All the changes for your comments in our manuscript have been highlighted by blue text. Here below we address the questions and suggestions.

No.	Comment	Response
1	The relation between tuberculosis and covid in the introduction should be added that may relate to post tuberculosis or severe tuberculosis infection that compromises the lung capacity.	A very brief review about relation between TB and COVID-19 has been added into the introduction. **Please Check for P4.** *There have been some studies suspected that the COVID-19 could reactivate or worsen TB. From the perspective of pathology, the disorders induced by SARS-COS-2 and Mycobacterium tuberculosis to the immunomodulation tend to induce an unbalanced inflammatory response, which can promote the progression and worsening of both diseases. Besides, the use of immunosuppressive drugs on severe COVID-19 patients may lead to increased likelihood of active TB caused by reactivation or new infection.*
2	The included studies were comprised of	These 2 different kinds of studies were assessed

	retrospective studies and case reports. How the evaluation of quality studies? good? fair? bad? Only good quality of study could advance to meta analysis.	by 2 different tools: Joanna Briggs Institute (JBI) Critical Appraisal Checklist for Study Reporting Prevalence Data and JBI Critical Appraisal Checklist for Case Report. **All the detail information is at P9.** *We used the Joanna Briggs Institute (JBI) Critical Appraisal Checklist for Study Reporting Prevalence Data for the prevalence part as the tool. Nine questions in this checklist were set to assess the quality of studies, including sample frame, sample size, data analysis, etc. All included studies were case reports for the treatment and outcome review, so we applied JBI Critical Appraisal Checklist for Case Report as the tool which consists of 8 questions about patients' condition, test and assessment, intervention, etc.* Based on the data we gathered, case reports and retrospective cohort studies provided 2 different kinds of information exclusively. The 35 case reports provided information about treatment and clinical outcomes; the 7 retrospective cohort studies provided information about fatality rate. As for the quality, I'm afraid the all JBI checklists (not just the 2 we used in this paper) are used to understand problems with bias and inform the approach to systematic review and meta-analysis. There is no way for us to score the questions or calculate indicators to decide on a cut-off to define high versus low quality. The quality of all included studies was summarized at section 3.4 Quality of Included Studies. We also discussed the possible bias at discussion part. I agree that good quality is very important for meta-analysis. While based on our assessment,

		most included retrospective cohort studies were suffered from insufficient sample size and inappropriate sample frame. I'm afraid that is all the evidences we have now, although their quality might not meet our expectation.
3	In the discussion should discuss the limited current evidence between Tuberculosis and COVID 19, further large prospective cohort studies are needed.	Yes, more studies, especially the experimental ones, are urgently need in this field. I have refined the manuscript. **Please Check for P21.** *Till to the searching date, the number of relevant studies were limited, cohort studies even more rare. Besides, we did not find any experimental study about TB-COVID coinfection. Considering the quality of current evidences, we think more studies are urgently needed to evaluate the prevalence, treatment, and long-term outcome of TB-COVID coinfection.*
4	The conclusion was to overestimate the results. The conclusion: there was still limited evidence to evaluate the association between tuberculosis and covid 19. Further prospective cohort studies are needed.	We have refined the discussion and conclusion part by softening the tone. We also added more contents about the limitation of current evidences as well as our study. **Please Check for P24-25.** *It's necessary to the results of MA were based on limited number of studies, of which defects in the quality was a frequent feature.* *Current evidence is insufficient to evaluate the treatment and outcome, not to estimate the prevalence rate of TB-COVID coinfection globally. More study about this field is in urgent need, especially experimental study and cohort study.*

VERSION 2 – REVIEW

REVIEWER	Wilkinson, Robert Wellcome Centre for Infectious Diseases Research in Africa
REVIEW RETURNED	19-Apr-2022

GENERAL COMMENTS	The authors have conscientiously and cordially responded to my comments